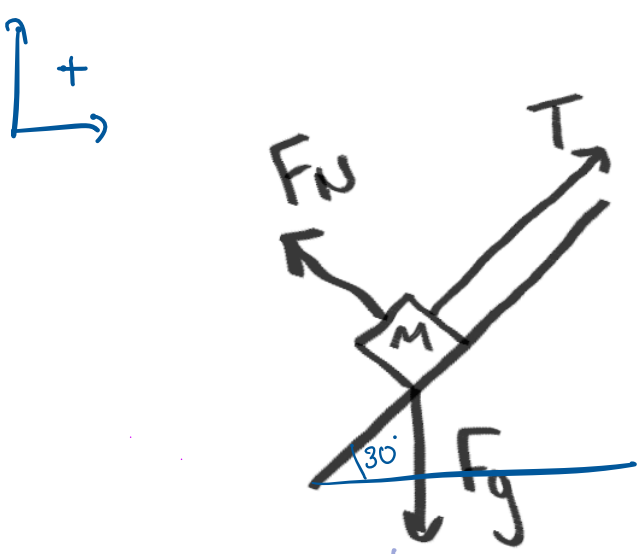
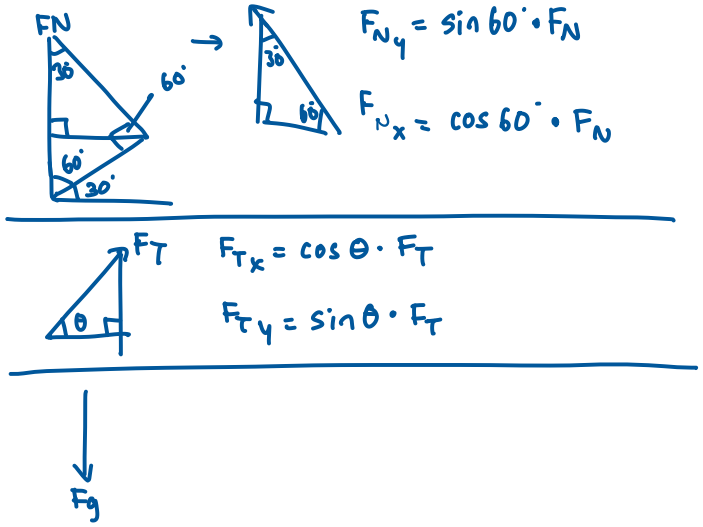
Changing frame of reference

October 3, 2026

1 Change reference frame

Suppose you have a mass M on an inclined slope, 30 degrees up from the horizon. The mass is supported in place by a string with Tensile Force T.





1. $\Sigma F_y = 0 = -F_g + \sin 60 \cdot F_N + \sin\theta \cdot F_T$

$\Sigma F_x = 0 = -\cos 60^\circ \cdot F_N + \cos\theta \cdot F_T$

1. Write out the two x and y equations of motion in the usual x-y coordinate system, knowing that the block is stationary.

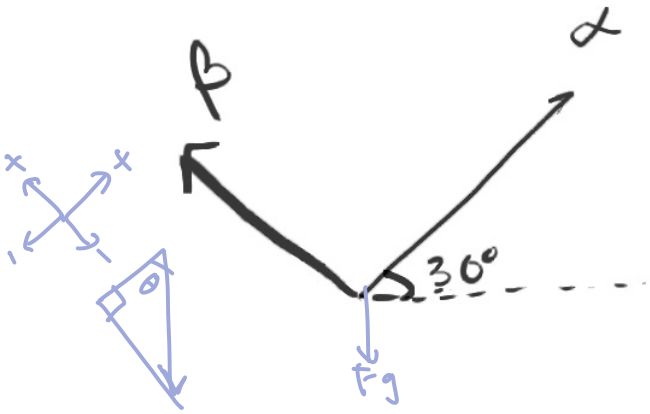
2. If only the mass were specified, say $M = 50kg$, would you be able to use just one equation to solve for either the tensile force or the normal force?

2. $F = ma$

No, because even if you set either the ΣF_y or ΣF_x to ma_x or ma_y, its not very helpful in one equation because $a = 0$ and there are still 2 uknowns.

You would need to use subtraction or substitution (2 equations)

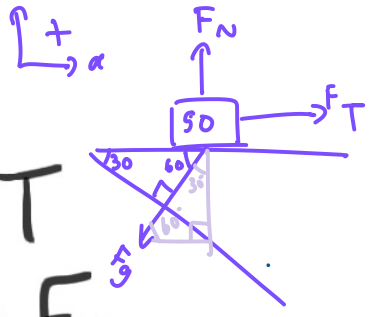
3. Consider this new coordinate system, with α replacing x and β replacing y.



$\Sigma F_B = 0 = F_N - \sin\theta \cdot F_g$

$T_\alpha = T$

$F_{N_\beta} = F_N$



① $F_g: 50 \cdot 9.8 = 490\ N$

② $\Sigma F_\beta = 0 = F_N - \sin 60 \cdot 490$

$0 = F_N - 424.35$

$424.35 = F_N$

③ $\Sigma F_\alpha = 0 = F_T - \cos 60 \cdot 490$

$245 = F_T$

Note that in this coordinate system, $T_\alpha = T$ because T is perfectly in line with the alpha axis. $F_{N\beta} = F_N$ because the normal force is perfectly in line with the beta axis. Rewrite the two equations of motion for the alpha and beta axes. For instance, $\Sigma F_{alpha} = 0 = T - F_{g\alpha}$. What is the β equation? $\Sigma F_\beta = 0 = F_N - \sin\theta \cdot F_g$

4. If only the mass were specified as $M = 50kg$ and you didn't know the tensile force or the normal force, solve for them.

$245\ N = F_T$

$424.35 = F_N$

2 BONUS

In either coordinate system we have two equations and two unknowns, which means we have enough information to completely solve it, but it wasn't as easy in the x-y coordinate system. Can you find an expression for F_N in the x-y coordinate system?

① $0 = -\cos 60 \cdot F_N + \cos 30\ F_T$

② $0 = -490 + \sin 60\ F_N + \sin 30\ F_T$

1. $F_T = \dfrac{+\cos 60\ F_N}{\cos 30}$
2. ① into ②
3. $490 = \sin 60\ F_N + \sin 30 \left(\dfrac{+\cos 60\ F_N}{\cos 30}\right)$